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FORMAT

- 10 multiple choice questions
- 1 question contains multiple correct answers
- 9 questions contain only one right answer
- Each answer is accompanied by feedback

QUIZ INTERPRETATION STEPS

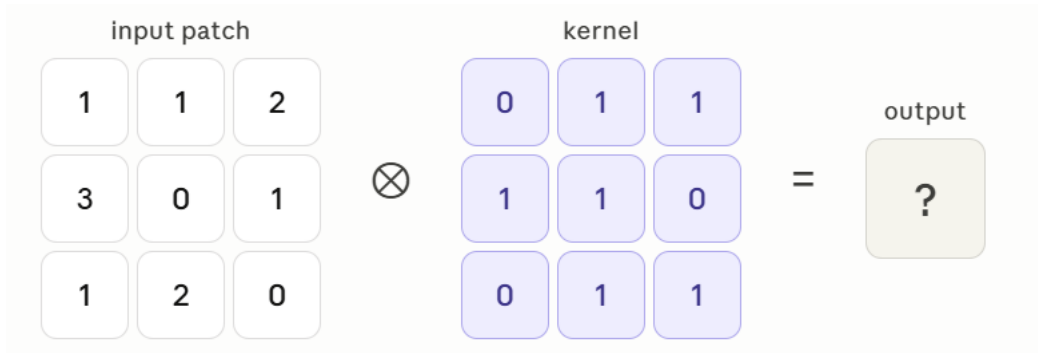
The use of this document is intended for learning purposes about the core fundamental aspects of **Convolutional Neural Networks** (CNNs). The ten questions address the basic aspects of Convolution, Spatial Arithmetic and Receptive Field. In preparation for taking this quiz, one must be familiar with the descriptions outlined in the accompanying study notes and simulator package.

Following a thorough understanding of the study notes and literature about this subject, one should refer to the link presented in this document and take the quiz. This is not a timed event, so it is advised to take as much time as necessary to evaluate the correct answer prior to submitting the final quiz. Upon completion, access the 'View Score' section to review the correctness of each result along with the feedback for each question. Evaluating the feedback, especially for wrong answers, helps to better understand the core concepts of this topic.

For further information about each question and the options available for selection, the next sections of this document present a thorough explanation. Note that this section shall be referenced only after submitting the quiz and evaluating the feedback given. Otherwise, the results submitted will be biased.

QUESTION 1 [one correct answer]

The basic convolution arithmetic consists of the kernel filtering. Determine the correct answer for the following example:



Answer A: [incorrect]

☐ 22

Feedback: This represents the addition of all elements from the resulting [3x3] matrix when performing normal multiplication [RxC] of the two matrices. This is a common pitfall when one is perceiving the multiplication of sliding kernels

Answer B: [correct]

☐ 8

Feedback: correct element-wise multiplication

Answer C: [incorrect]

☐ 18

Feedback: This total score represents the product of the sum of element-wise multiplication of each row. This is not correct as the total of each row should be added not multiplied.

Answer D: [incorrect]

☐ 6

Feedback: While we do element-wise multiplication each row should be multiplied with the corresponding row not the column

QUESTION 2 [one correct answer]

For a **Kernel** of size ' k ', an input of size ' i ' and padding and stride of sizes ' p ' and ' s ' respectively what is the general formula representing output size ' o '?

Answer A: incorrect]

☐ $o = i + 2p - k + 1$

Feedback: Does not take into consideration stride effect ' s '

Answer B: [incorrect]

$$[] o = \left(\frac{i+2p-k}{s} \right) + 1$$

Feedback: The elements are correctly computed but the brackets do not represent the flooring element. The latter ensures that the output is always an integer value representing the dimension of the output.

Answer C: [correct]

$$[] o = \left\lfloor \frac{i+2p-k}{s} \right\rfloor + 1$$

Feedback: correct answer.

Answer D: [incorrect]

$$[] o = \left\lceil \frac{i+2p-k}{s} + 1 \right\rceil$$

Feedback: Returns same correct result but is not accurately written

QUESTION 3 [one correct answer]

What is 'full padding' in the context of convolution arithmetic, and what is the resulting output size 'o' when stride 's' = 1 ?

Answer A: [incorrect]

$$[] p = 0, \text{ resulting in } o = i - k + 1$$

Feedback: $p=0$ means no padding at all is considered. This is sometimes also known as no padding.

Answer B: [correct]

$$[] p = k - 1, \text{ resulting in } o = i + (k - 1)$$

Feedback: correct answer as it allows partial superimposition while expanding output beyond input

Answer C: [incorrect]

$$[] p = \lfloor k/2 \rfloor, \text{ resulting in } o = i$$

Feedback: This represents half padding rather than full padding

Answer D: [incorrect]

$$[] p = k, \text{ resulting in } o = i + k$$

Feedback: if $p = k$ then it will result in positions where kernel and input do not overlap

QUESTION 4 [one correct answer]

How does a stride greater than 1 in an early layer affect the total receptive field size (r_o) of a deep network?

Answer A: [incorrect]

☐ It has a subtractive effect based on the padding size.

Feedback: While stride is a form of sub-sampling it does not reduce the RF. Alternatively it increases the receptive field.

Answer B: [incorrect]

☐ It has no effect on the receptive field; it only affects output dimensions.

Feedback: while stride reduced the output dimension it increases the receptive field.

Answer C: [incorrect]

☐ It has a linear effect, adding to the total size.

Feedback: Wrong. It has a multiplicative effect on the RF

Answer D: [correct]

☐ It has a multiplicative effect, proportionally increasing the region covered for all subsequent layers.

Feedback: If the stride is greater than 1 for a particular layer, the receptive field region increases proportionally for all layers below that one. Thus, the end effect is the product of each stride value at each layer.

QUESTION 5 [one correct answer]

How does a **dilation** of rate 'd' effect the kernel size of original size 'k' into an effective kernel ' $\hat{k}$ '?

Answer A: [incorrect]

☐ $\hat{k} = k + d$

Feedback: A linear increase in the kernel does not support the scope

Answer B: [correct]

☐ $\hat{k} = d(k - 1) + 1$

Feedback: correct answer

Answer C: [incorrect]

☐ $\hat{k} = k \cdot d$

Feedback: fails to account for the end points of the kernel

Answer D: [incorrect]

$$[\] \hat{k} = k(d - 1) + 1$$

Feedback: mathematic incorrect interpretation as the elements 'k' and 'd' are swapped

QUESTION 6 [one correct answer]

In a single-path network, if you have the receptive field size ' r_l ' at layer , what is the formula to determine the receptive field size at the previous layer?

Answer A: [incorrect]

$$[\] r_{l-1} = s_l(r_l - 1) + k_l$$

Feedback: not correct

Answer B: [incorrect]

$$[\] r_{l-1} = r_l + (k_l - 1)$$

Feedback: not correct. Only valid when stride = 1 and does not consider the compounded effect of multiple layers

Answer C: [correct]

$$[\] r_{l-1} = s_l \cdot r_l + (k_l - s_l)$$

Feedback: Correct. Accounts for scaling across multiple layers and the spatial context of the kernel size. Alternatively, $\hat{k} = k + (k-1)(d-1)$

Answer D: [incorrect]

$$[\] r_{l-1} = s_l + r_l + (k_l - 1)$$

Feedback: Addition of stride effect to the receptive field is not correct as the product shall be considered.

QUESTION 7 [one correct answer]

The adjusted output equation when considering a dilated convolution of rate, d , is represented by 'o' as with the following equation:

Answer A: [incorrect]

$$[\] o = \left\lfloor \frac{i+2p-k}{s} \right\rfloor + 1$$

Feedback: This formula does not take into consideration dilation effect

Answer B: [incorrect]

$$[\] o = \left\lfloor \frac{i+2p-(k.d)}{s} \right\rfloor + 1$$

Feedback: Does not take into account the end point of the kernel

Answer C: [correct]

$$[] o = \left\lfloor \frac{i+2p-[d(k-1)+1]}{s} \right\rfloor + 1$$

Feedback: In a dilated convolution, the hyperparameter "inflates" the kernel by inserting spaces between its elements. This increases the effective kernel size ($\hat{k}$) to $d(k - 1) + 1$ (also expressed as $k + (k - 1)(d - 1)$). To find the resulting output dimension, this inflated size must be substituted for the original kernel size (k) in the standard convolution arithmetic formula in question 2

Answer D: [incorrect]

$$[] o = \left\lfloor \frac{i+2p-k}{s.d} \right\rfloor + 1$$

Feedback: Dilation has a multiplicative impact on the original kernel. Cannot form part of the denominator

QUESTION 8 [multiple correct answer]

Which of the following are significant limitations of standard, fully connected Artificial Neural Networks (ANNs) when applied to image recognition, as compared to Convolutional Neural Networks (CNNs)? (Select all that apply)

Answer A: [correct]

☐ Their global connectivity means every unit in a hidden layer depends on the entire input, failing to exploit the strong local correlations inherent in visual data.

Feedback: CNNs restrict unit connections to local receptive fields to extract elementary visual features like edges or corners, whereas ANNs suffer from being "unstructured" and globally connected.

Answer B: [incorrect]

☐ They are unable to learn complex, high-dimensional mappings from examples, making them theoretically inferior for any non-linear classification task.

Feedback: Both ANNs and CNNs are able to learn complex mappings

Answer C: [correct]

☐ They possess an excessive number of parameters when processing high-resolution images, which increases system capacity and requires much larger training sets to prevent overfitting.

Feedback: Large images lead to a "prohibitive" number of weights in fully connected layers (e.g., tens of thousands for just 100 hidden units), which increases the capacity and the amount of data needed for training.

Answer D: [correct]

☐ They entirely ignore the spatial topology of the input image, treating pixels as an unordered collection of features where the variables could be presented in any order without affecting the training outcome.

Feedback: Fully connected architectures ignore the 2D local structure of images, whereas CNNs use local connectivity to exploit the fact that nearby pixels are highly correlated. This makes them sensitive to small shifts in data

QUESTION 9 [one correct answer]

The 'Effective stride' S_0 of a complete network with L layers is represented as:

Answer A: [incorrect]

☐ $S_0 = \max(s_1, s_2, s_3, \dots, s_L)$

Feedback: This does not take into consideration the cumulative effect across all layers

Answer B: [correct]

☐ $S_0 = \prod_{l=1}^L s_l$

Feedback: Stride effect is the product of the individual strides of each layer

Answer C: [incorrect]

☐ $S_0 = \sum_{l=1}^L s_l$

Feedback: Stride effect is not the additive of the individual strides of each layer

Answer D: [incorrect]

☐ $S_0 = \sum_{l=1}^L (1/s_l)$

Feedback: Stride effect is the product of each stride value at the various layers and should not be mistaken by the denominator effect on the output

QUESTION 10 [one correct answer]

Regarding the evolution of the Effective Receptive Field (ERF) during the training process of a CNN, what has been empirically observed?

Answer A: [incorrect]

☐ ERF remains constant throughout the training process as it is a fixed element of the architecture

Feedback: The concept of ERF derives from the research findings that as weights and parameters are being trained the ERF changes significantly

Answer B: [incorrect]

☐ ERF shrinks during training to focus on local features

Feedback: A shrinking ERF during training will defy the purpose of model training

Answer C: [correct]

☐ ERF typical starts small and grows during the learning cycles

Feedback: studies show that when random weights are assigned ERF is small and it expands significantly as model reaches higher accuracy
